

Exam01 Practice Questions

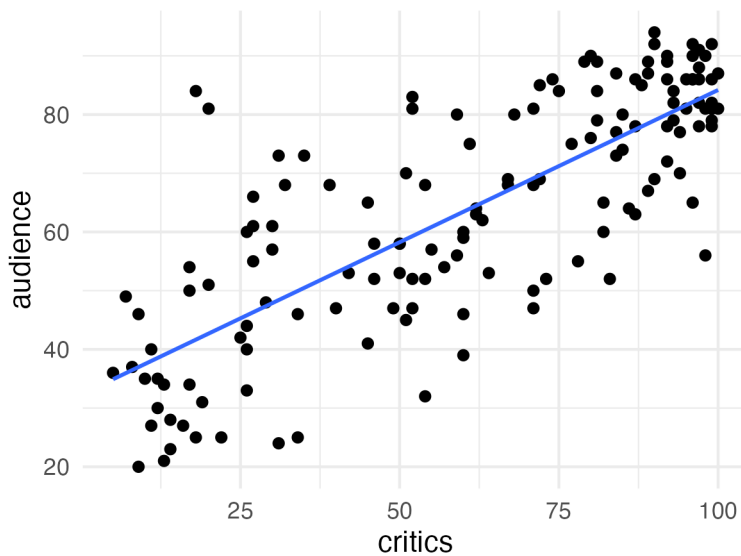
Stat 230 - Spring 2026

Practice questions for each skill on the [Exam01 Review](#) sheet.

1. Express a SLR model using proper notation

Example question In the Day 2 activity, we explored the relationship between **audience** score and **critics** score for a sample of movies. Here is the scatterplot we looked at:

```
movies <- read.csv("https://aloy.github.io/stat230-materials/data/movie_scores.csv")
gf_point(audience ~ critics, data = movies) |>
  gf_lm()
```



(a) What is the response variable? What is the explanatory variable?

- (b) Write out the *theoretical* SLR model for this relationship. Define Y_i and x_i in context, include all parameters, and state the assumption placed on the error term.

Example question The Day 3 activity fit a model relating leaf width (**Width**, in mm) to **Year** for a sample of 252 leaves, giving the coefficient estimates below.

term	estimate
(Intercept)	37.723
Year	-0.018

Write out the *fitted* SLR model using proper notation.

2. Interpret the coefficients and R^2 of a model in context

```
LeafWidth <- read_csv("https://aluby.github.io/stat230/data/LeafWidth.csv")
leaf_mod <- lm(Width ~ Year, data = LeafWidth)
```

Example question Using the fitted model ($\widehat{\text{Width}}_i = 37.723 - 0.018 \times \text{Year}_i$), interpret the slope and intercept in context.

Example question For the **LeafWidth** model above, predicting **Width** from **Year** has $R^2 = 0.06$. Which of the following is the best interpretation of this value?

- (a) For each additional year, leaf width increases by 0.6mm on average.
- (b) About 6% of leaves have their width predicted exactly correctly by this model.
- (c) About 6% of the variability in width among these leaves is explained by year.
- (d) The correlation between width and year is 0.6.

3. Check necessary conditions for a regression model and discuss the implications of a model deficiency

Example question For each statement below, circle TRUE or FALSE.

- (a) TRUE or FALSE: We can check the linearity condition using a scatterplot of the response against the explanatory variable, before ever fitting a model.
- (b) TRUE or FALSE: A normal Q-Q plot that curves away from the diagonal line in both tails suggests the residuals are more heavily-tailed than a Normal distribution.
- (c) TRUE or FALSE: A residuals-vs-fitted-values plot that fans out (wider spread for larger fitted values) suggests a violation of the equal variance condition.

- (d) TRUE or FALSE: If the independence condition is violated, we can typically detect it just by looking at a residuals-vs-fitted plot.
- (e) TRUE or FALSE: If the linearity condition is badly violated, the estimated slope may be misleading, even if the p-value for the slope is small.

Example question Suppose you fit a model predicting **income** from **education** for the Riverview data, and the residuals-vs-fitted plot shows a clear curved (U-shaped) pattern. Which of the following can you still trust about this model? Select all that apply, and explain your reasoning.

- (a) The point estimate of the slope
- (b) The p-value for the slope
- (c) The standard error of the slope
- (d) Predictions made from the model

4. Perform a hypothesis test to draw inference about a single regression coefficient

Example question The LeafWidth model gives the following output:

```
broom::tidy(leaf_mod) |>
  knitr::kable(digits = 3)
```

term	estimate	std.error	statistic	p.value
(Intercept)	37.723	8.575	4.399	0
Year	-0.018	0.004	-4.029	0

On $n = 252$ data points. Conduct a hypothesis test for $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$. State the test statistic, and describe (without necessarily calculating exactly) how the p-value could be found.

Example question Before collecting the NHANES sitting-time data, a researcher hypothesizes that *more* sitting time is associated with a *higher* depression risk score – not just a difference in either direction. The fitted model (`dep_score ~ sedentary_hours`) gives a slope of 0.0314 (SE = 0.0116) for `sedentary_hours`, with $df = 657$.

- (a) State H_0 and H_A in symbols, matching the researcher's directional hypothesis.
- (b) Is the p-value R reports by default (from `summary()`) the correct one to use here? If not, explain how you would find the correct p-value.

5. Construct and interpret confidence intervals for regression coefficients

Example question Recall Movies model: `lm(audience ~ critics)` gives a slope estimate of 0.519 (SE = 0.0345), with $df = 144$.

- (a) Using the `qt()` output below, report the correct t^* value to use rounded to 3 decimal places

```
qt(.01, df = 144, lower.tail = FALSE)
```

```
[1] 2.352522
```

```
qt(.025, df = 144, lower.tail = FALSE)
```

```
[1] 1.976575
```

```
qt(.05, df = 144, lower.tail = FALSE)
```

```
[1] 1.655504
```

```
qt(.1, df = 144, lower.tail = FALSE)
```

```
[1] 1.287458
```

- (b) Construct and report a 90% CI for the slope, showing the formula with numbers plugged in.

Example question Recall Riverview model using *standardized* education: `income ~ scale(education)` gives a slope estimate of 11.567 (SE = 1.613), with $df = 30$. For a 95% confidence interval, use a critical value is $t^* = 2$.

It may also be helpful know that the `income` has a mean of 53.7 and a SD of 14.5. `education` has a mean of 16 and a SD of 4.36.

- (a) Construct a 95% CI for this slope.
(b) Interpret the interval in context.

6. Use a regression model to make appropriate predictions and communicate the uncertainty of these predictions through an interval

Example question Using the movies model, we predict the audience score for a movie with a critics score of 75. The point estimate is 71.2, and R produces two 90% intervals: (50.4, 92.1) and (69.3, 73.1).

- (a) Which interval is the confidence interval, and which is the prediction interval? How can you tell?
- (b) If we wanted to predict the audience score for *Kingsman: The Secret Service*, which has a critics score of 75, which interval should we use?
- (c) For the type of interval you did *not* choose in (b), what is the correct interpretation?

Example question A researcher wants to use the NHANES `dep_score ~ sedentary_hours` model to make a claim about people who sit 6 hours per day. Explain which interval (confidence or prediction) is appropriate for each scenario below, and briefly say why.

- (a) “I want a range for the depression risk score of one specific person who sits 6 hours a day.”
- (b) “I want a range for the *average* depression risk score among everyone who sits 6 hours a day.”

7. Express an SLR model when transformations have been used on the response and/or the explanatory variables

Example question Instead of the raw PHQ-9 depression score, we actually used `dep_score`, defined as $\log(\text{PHQ-9} + 1)$. The fitted model is `dep_score ~ sedentary_hours`, with intercept 0.904 and slope 0.0314.

- (a) Write out the fitted SLR equation, being careful to write it in terms of `dep_score`, not the raw PHQ-9 score.
- (b) A student concludes: “for every one-hour increase in sitting time, the model predicts an average increase of 0.0314 points in PHQ-9 score.” Explain in one sentence why this is incorrect, then give a corrected interpretation.

Example question Suppose instead of transforming the response, we wanted to write out the general *theoretical* model for a log-transformed response, where Y_i is the raw response and x_i is the explanatory variable.

- (a) Write the theoretical model using proper notation, and state the assumption on the error term.
- (b) Give one reason (from what we discussed in class) why we might choose to log-transform a response variable before fitting a model.